Answer Key — Lenses

Final answer with a one-line reason. Full methods are in the Notes' worked examples.

Objective

Q1 **(b) thicker in the middle** — a convex lens is thicker in the middle.

Q2 **(b) converging** — a convex lens converges light.

Q3 **(b) convex lens** — a magnifying glass is a convex lens.

Q4 **(b) lets light pass through it** — lenses transmit light; we see through them.

Q5 **(a) Both true, R explains A** — diverging light gives a small erect image — as R explains.

Q6 **concave** — thick edges → concave lens.

Q7 **converges** — a convex lens converges light.

Short answer

Q8 Convex: thicker in the middle, converges light. Concave: thicker at the edges, diverges light.

Q9 A mirror reflects light and we see images in it; a lens is transparent and lets light pass through, so we see through it.

Q10 Convex: magnifying glass / spectacles for long sight. Concave: spectacles for short sight (any one each).

Long / application

Q11 The water drop has an outward-curved surface, so it behaves like a convex lens. It converges light from the text, forming an enlarged, erect image when the text is close — just like a magnifying glass. So the letters look bigger.

Q12 A convex mirror reflects light and diverges a parallel beam; we see an erect, diminished image *in* it. A convex lens transmits light and converges a parallel beam; we see images *through* it (enlarged and erect for a close object). So one reflects and diverges, the other transmits and converges.

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